

Name: _____ Date: _____

Period: _____

Parametrization Practice — Key

1. $\frac{x^2}{9} + \frac{y^2}{25} = 1$

(a) $x(t) = 3 \cos t$ $y(t) = 5 \sin t$

(b) Extreme points:

Extreme point	t	Coordinates
Rightmost	0	(3, 0)
Leftmost	π	(-3, 0)
Topmost	$\pi/2$	(0, 5)
Bottommost	$3\pi/2$	(0, -5)

2. $\frac{(x-3)^2}{16} + \frac{(y+1)^2}{4} = 1$

(a) $h = 3, k = -1, a = 4, b = 2$. $x(t) = 3 + 4 \cos t$, $y(t) = -1 + 2 \sin t$, $0 \leq t \leq 2\pi$.

(b) **Set $y(t) = 0$:** $-1 + 2 \sin t = 0 \Rightarrow \sin t = \frac{1}{2} \Rightarrow t = \pi/6$ or $t = 5\pi/6$.

At $t = \pi/6$: $x = 3 + 4 \cos(\pi/6) = 3 + 4 \cdot \frac{\sqrt{3}}{2} = 3 + 2\sqrt{3} \approx 6.46$.

At $t = 5\pi/6$: $x = 3 + 4 \cos(5\pi/6) = 3 - 2\sqrt{3} \approx -0.46$.

x -axis crossings: $(3 + 2\sqrt{3}, 0)$ and $(3 - 2\sqrt{3}, 0)$.

Parabola Parametrization — Key

3. $y = 2(x-1)^2 - 3$; $x(t) = 1 + t$, $y(t) = 2t^2 - 3$.

(a) **Set $y(t) = 0$:** $2t^2 - 3 = 0 \Rightarrow t^2 = \frac{3}{2} \Rightarrow t = \pm \sqrt{\frac{3}{2}} = \pm \frac{\sqrt{6}}{2}$.

$x = 1 + t$: $x = 1 + \frac{\sqrt{6}}{2}$ and $x = 1 - \frac{\sqrt{6}}{2}$.

x -intercepts: $(1 \pm \frac{\sqrt{6}}{2}, 0)$.

(b) **Vertex is the minimum of $y = 2(x-1)^2 - 3$, which is $(1, -3)$.** **At $t = 0$:** $x(0) = 1 + 0 = 1$, $y(0) = 2(0)^2 - 3 = -3$. **Confirmed:** $(1, -3)$.

4. $x^2 + y^2 = 49$

(a) $x(t) = 7 \cos t$ $y(t) = 7 \sin t$

(b) $x(\pi/3) = 7 \cos(\pi/3) = 7 \cdot \frac{1}{2} = \frac{7}{2}$. $y(\pi/3) = 7 \sin(\pi/3) = 7 \cdot \frac{\sqrt{3}}{2} = \frac{7\sqrt{3}}{2}$. **Coordinates:** $(\frac{7}{2}, \frac{7\sqrt{3}}{2})$.

AP-Style Practice — Key

5. Correct answer: **(B)** $x(t) = 5 \cos t$, $y(t) = 3 \sin t$.

Teacher Note

In $\frac{x^2}{25} + \frac{y^2}{9} = 1$, the denominators are $a^2 = 25$ and $b^2 = 9$, so $a = 5$ and $b = 3$. The parametrization is $x(t) = a \cos t = 5 \cos t$, $y(t) = b \sin t = 3 \sin t$. (A) swaps a and b . (C) uses a^2 and b^2 instead of a and b — a very common error; emphasize that students take the square root of the denominator. (D) also swaps a and b .